
SOFTWARE REQUIREMENTS SPECIFICATION

For

Chinese Dragon Restaurant Web Application

Version 1.0 approved

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1. INTRODUCTION

1.1 PURPOSE

This document specifies the software requirements for the **Chinese Dragon Restaurant Web Application** (Version 1.0). The system allows customers to browse menu items, place orders, submit reviews, and contact the restaurant. Administrators can manage menu items, view all orders, and respond to contact messages. This SRS serves as a guideline for design, development, testing, and documentation.

1.2 DOCUMENT CONVENTIONS

- Requirements are labeled as **REQ-1, REQ-2, ...**
- High-level headings indicate sections; subheadings indicate specific topics.
- Priority is indicated as **High, Medium, or Low.**

1.3 INTENDED AUDIENCE AND READING SUGGESTIONS

- **Developers:** Refer to Sections 2–4 for detailed functional requirements and design considerations.
- **Testers:** Use Sections 4–5 for system testing coverage.
- **Stakeholders:** Overview sections 1–2 and 4 provide functional understanding.

1.4 PRODUCT SCOPE

The software provides a **responsive web application** for ordering from Chinese Dragon Restaurant. Customers can view menu items, place orders, submit reviews after completed orders, and contact the restaurant. Admin can manage menu items, view orders, and view customer messages.

1.5 REFERENCES

- IEEE 830-1998 Standard for Software Requirements Specification
- MERN Stack Documentation (MongoDB, Express.js, React.js, Node.js)
- bcrypt library documentation for password encryption

2. OVERALL DESCRIPTION

2.1 PRODUCT PERSPECTIVE

- Standalone web-based system using **MERN stack**.
- Frontend: React.js (SPA, responsive design).
- Backend: Node.js with Express.js (REST API).
- Database: MongoDB.

2.2 PRODUCT FUNCTIONS

- User authentication (register, login, logout)
- Browse menu with descriptions, price, and portion size
- Place orders and view order history
- Submit reviews for completed orders
- Contact form for customer inquiries
- Admin: Manage menu and view orders/messages

2.3 USER CLASSES AND CHARACTERISTICS

User Type	Capabilities	Requirements
Customer	Browse menu, place orders, submit reviews, contact restaurant	Basic internet knowledge
Admin	Manage menu, view orders/messages	Authorized access only

2.4 OPERATING ENVIRONMENT

- Web browsers: Chrome, Firefox, Edge, Safari
- Node.js runtime, MongoDB server

2.5 DESIGN AND IMPLEMENTATION CONSTRAINTS

- Must use MERN stack
- Secure authentication using JWT
- Reviews only from verified customers with completed orders
- No order tracking required
- Responsive design

2.6 USER DOCUMENTATION

- Online help for customers
- Admin manual for menu/order management

2.7 ASSUMPTIONS AND DEPENDENCIES

- Valid customer data
- Admin maintains system data
- Email server optional for contact responses

3. EXTERNAL INTERFACE REQUIREMENTS

3.1 USER INTERFACES

Customer: Home, Menu, Cart/Order, Contact Us, Review Submission

Admin: Menu Management, Orders View, Contact Messages

3.2 HARDWARE INTERFACES

Standard desktop/mobile devices with internet access

3.3 SOFTWARE INTERFACES

- MongoDB
- Node.js + Express backend
- React.js frontend SPA

3.4 COMMUNICATIONS INTERFACES

- HTTP/HTTPS
- JSON for API communication

4. SYSTEM FEATURES

4.1 USER AUTHENTICATION

Description & Priority: Login, registration, logout; **High** priority

Stimulus/Response:

- Customer registers → system validates → success or error message
- Customer logs in → system authenticates → access granted

Functional Requirements:

- REQ-1: Register with name, email, password
- REQ-2: Login with email and password
- REQ-3: Logout
- REQ-4: Passwords encrypted using bcrypt
- REQ-5: Validate email uniqueness

4.2 MENU MANAGEMENT

Description & Priority: Display/manage menu; **High** priority

Functional Requirements:

- REQ-6: Display all menu items
- REQ-7: Menu item fields: name, category, description, price, portion size
- REQ-8: Admin can add/update/delete menu items

4.3 ORDER MANAGEMENT

Description & Priority: Place and manage orders; **High** priority

Functional Requirements:

- REQ-9: Add items to cart
- REQ-10: Place orders
- REQ-11: Store orders in database
- REQ-12: Customers can view past orders
- REQ-13: Admin can view all orders
- REQ-14: Admin can mark orders as completed

4.4 REVIEW SYSTEM

Description & Priority: Submit reviews for completed orders; **Medium** priority

Functional Requirements:

- REQ-15: Only verified customers with completed orders can submit reviews
- REQ-16: Reviews include rating (1–5) and comment
- REQ-17: Reviews displayed on homepage

4.5 CONTACT US SYSTEM

Description & Priority: Customers contact restaurant; **Medium** priority

Functional Requirements:

- REQ-18: Submit inquiries with name, email, and message
- REQ-19: Admin can view messages

5. OTHER NONFUNCTIONAL REQUIREMENTS

5.1 PERFORMANCE REQUIREMENTS

- Page load < 3 seconds
- Handle multiple concurrent users

5.2 SAFETY REQUIREMENTS

- Prevent data loss with backup

5.3 SECURITY REQUIREMENTS

- JWT authentication
- Password hashing
- Input validation to prevent injection

5.4 SOFTWARE QUALITY ATTRIBUTES

- Responsive UI
- Modular, maintainable code

5.5 BUSINESS RULES

- Only verified customers can review orders
- Admin manages menu/orders/messages

6. OTHER REQUIREMENTS

- Future: Payment gateway, mobile app, analytics

APPENDIX A: GLOSSARY

- **MERN:** MongoDB, Express.js, React.js, Node.js
 - **JWT:** JSON Web Token
 - **Admin:** System administrator
 - **Customer:** Registered user
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APPENDIX B: ANALYSIS MODELS

- ER Diagram: Users, MenuItems, Orders, Reviews, ContactMessages

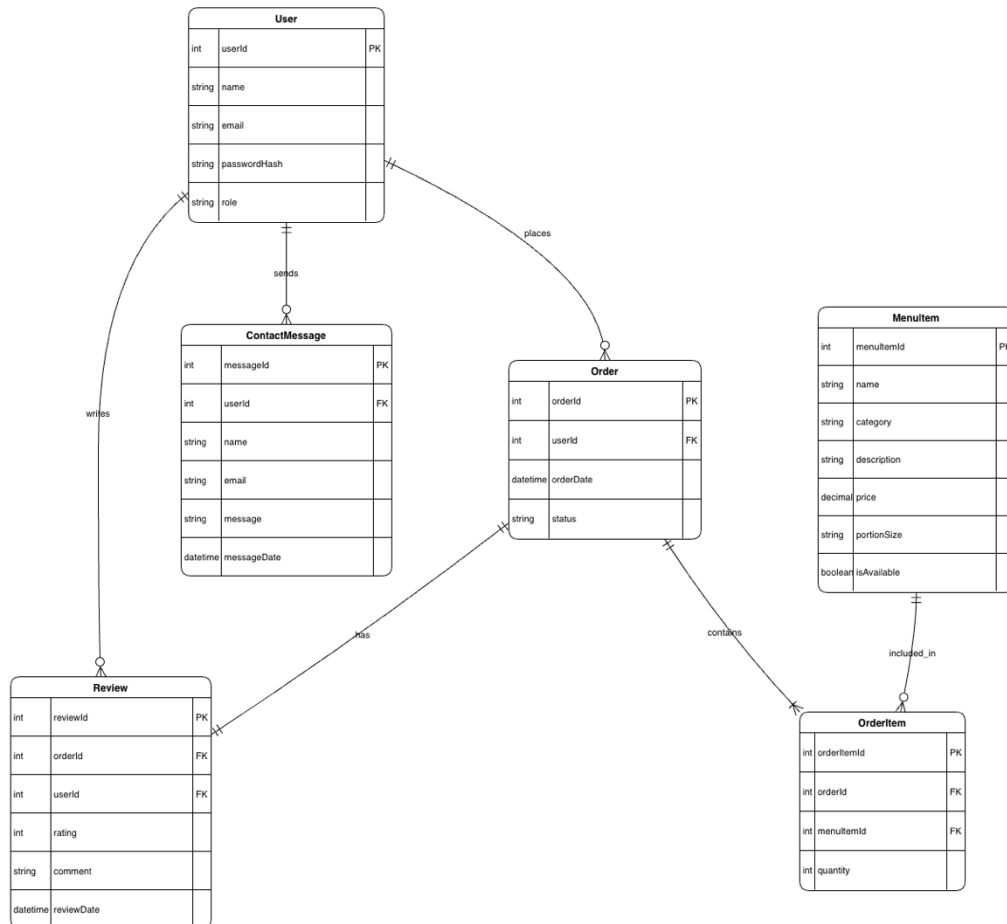


Figure B-1: Entity-Relationship Diagram for Chinese Dragon Restaurant Web Application

- Use Case Diagram: Customer & Admin → Place Order, Add Review, Manage Menu, Submit Contact

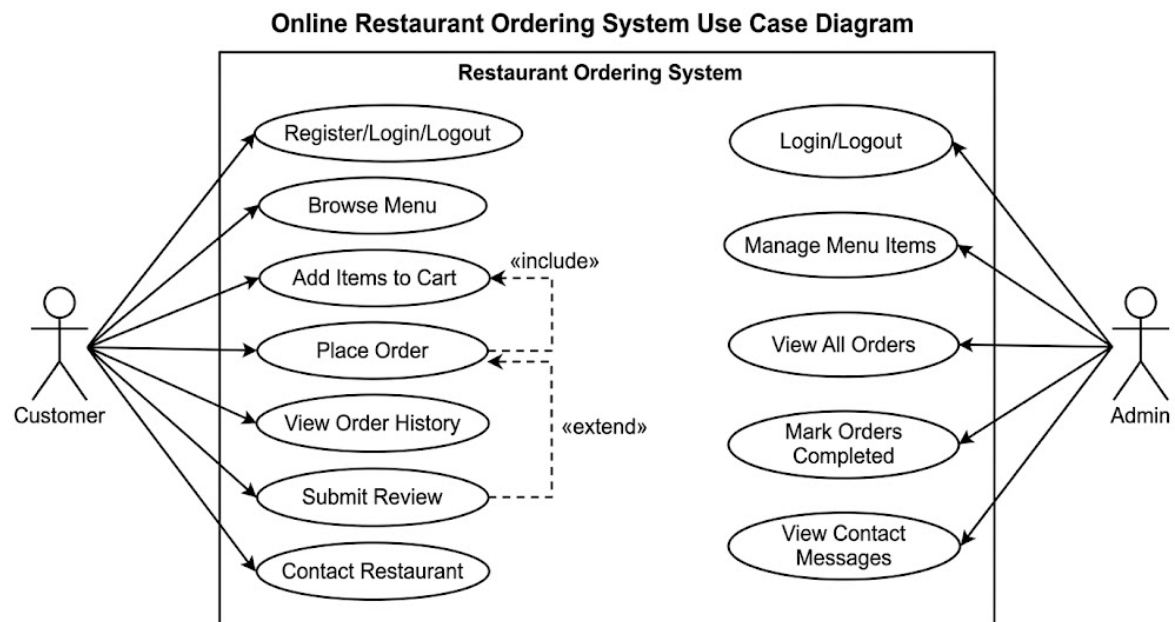


Figure B-2: Use Case Diagram for Chinese Dragon Restaurant Web Application

APPENDIX C: TO BE DETERMINED LIST

- Optional email notification system
- Future analytics dashboard